Experiment -1

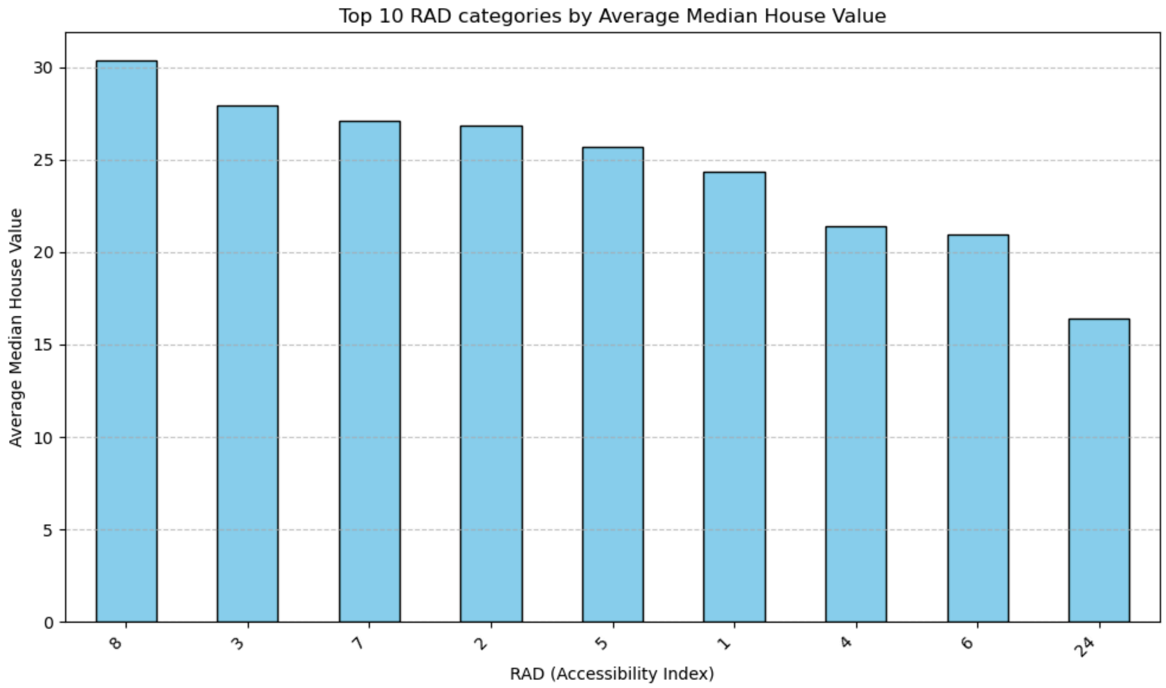
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Read the dataset
df = pd.read_csv("housing_data.csv")

# Experiment 1A: RAD vs MEDV
data = (df.groupby("RAD")["MEDV"]
        .mean()
        .sort_values(ascending=False))



# Plot
data.plot(kind="bar", color="skyblue", edgecolor="black", figsize=(10, 6))
plt.title("Average Median House Value by RAD")
plt.xlabel("RAD (Accessibility Index)")
plt.ylabel("Average Median House Value")
plt.xticks(rotation=45, ha="right")
plt.grid(axis="y", linestyle="--", alpha=0.7)
plt.tight_layout()
plt.show()
```

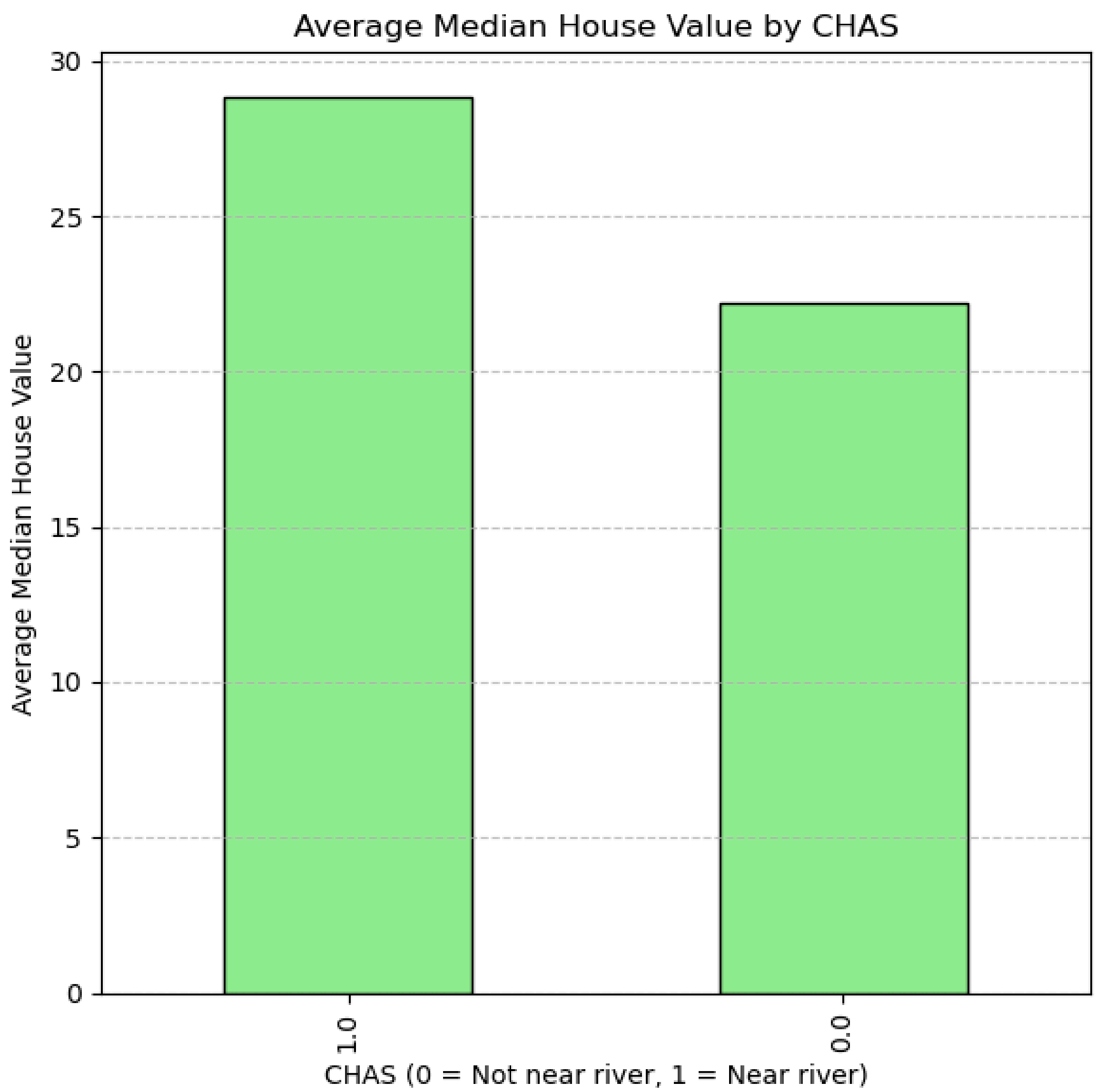


 # Experiment 1B:CHAS vs MEDV:-

```
data = (df.groupby("CHAS")["MEDV"]
        .mean()
        .sort_values(ascending=False))
```

#Plot

```
data.plot(kind="bar", color="lightgreen", edgecolor="black", figsize=(6, 6))
plt.title("Average Median House Value by CHAS")
plt.xlabel("CHAS (0 = Not near river, 1 = Near river)")
plt.ylabel("Average Median House Value")
plt.grid(axis="y", linestyle="--", alpha=0.7)
plt.tight_layout()
plt.show()
```



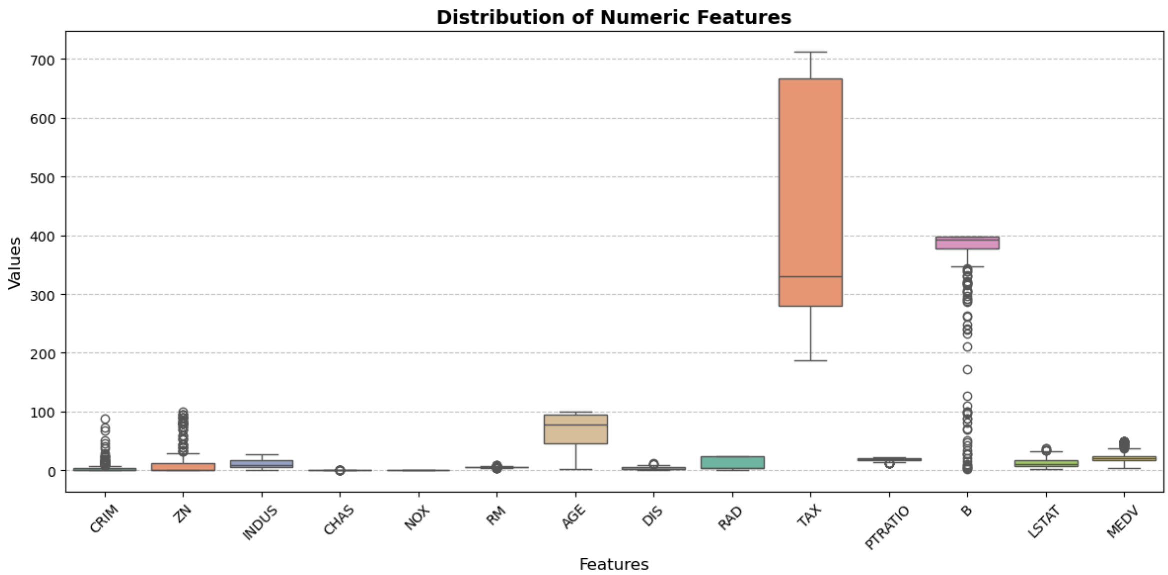
Experiment -2

 # Boxplot of numeric features

df_clean = df.replace([np.inf, -np.inf], np.nan).dropna()

#Plot

```
plt.figure(figsize=(12,6))
sns.boxplot(data=df_clean.select_dtypes(include='number'), palette="Set2")
plt.title("Distribution of Numeric Features", fontsize=14, fontweight="bold")
plt.xlabel("Features", fontsize=12)
plt.ylabel("Values", fontsize=12)
plt.xticks(rotation=45)
plt.grid(axis="y", linestyle="--", alpha=0.7)
plt.tight_layout()
plt.show()
```



Experiment -3

#Violin plot of numeric distributions

```
numeric_cols = df.select_dtypes(include='number').columns.tolist()
df_melt = df.melt(value_vars=numeric_cols,
            var_name='Feature',
            value_name='Value')
```

#Plot

```
plt.figure(figsize=(12,6))
sns.violinplot(data=df_melt, x='Feature', y='Value', palette="coolwarm")
plt.title("Distribution of Numeric Features", fontsize=16, fontweight="bold")
plt.xlabel("Feature", fontsize=12)
plt.ylabel("Value", fontsize=12)
plt.xticks(rotation=45)
plt.grid(axis="y", linestyle="--", alpha=0.6)
plt.tight_layout()
plt.show()
```